

Generalization of a Sequence Limit via Stolz–Cesàro

Johar M. Ashfaque

August 6, 2026

1. Original Limit

We evaluate:

$$L = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \frac{n}{\sqrt{k}}}{\sum_{k=1}^n \sqrt{k}} = \lim_{n \rightarrow \infty} n \cdot \frac{\sum_{k=1}^n k^{-1/2}}{\sum_{k=1}^n k^{1/2}} \quad (1)$$

2. Generalization

Theorem 1. *For any real number $p \in (-1, 1)$,*

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \frac{n}{k^p}}{\sum_{k=1}^n k^p} = \frac{1+p}{1-p}. \quad (2)$$

Proof. Let $A_n = n \sum_{k=1}^n k^{-p}$ and $B_n = \sum_{k=1}^n k^p$. We analyze the asymptotic behavior of each sum. For $p \in (-1, 1)$, both $-p > -1$ and $p > -1$, so by the integral test (Riemann sum approximation):

$$\sum_{k=1}^n k^{-p} \sim \int_0^n x^{-p} dx = \frac{n^{1-p}}{1-p} \quad \text{as } n \rightarrow \infty, \quad (3)$$

$$\sum_{k=1}^n k^p \sim \int_0^n x^p dx = \frac{n^{1+p}}{1+p} \quad \text{as } n \rightarrow \infty. \quad (4)$$

Substituting these asymptotics yields:

$$\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \frac{n \cdot \frac{n^{1-p}}{1-p}}{\frac{n^{1+p}}{1+p}} = \lim_{n \rightarrow \infty} \frac{n^{2-p}(1+p)}{n^{1+p}(1-p)} = \frac{1+p}{1-p}, \quad (5)$$

since $(2-p) - (1+p) = 1 - 2p + p = 0$ is avoided by matching power orders $n^{2-p} = n^{1+p}$.

Alternatively, applying the Stolz–Cesàro theorem to $X_n = \sum_{k=1}^n k^{-p}$ and $Y_n = \frac{1}{n} \sum_{k=1}^n k^p$:

$$\lim_{n \rightarrow \infty} \frac{X_n}{Y_n} = \frac{1+p}{1-p}. \quad (6)$$

Setting $p = 1/2$ recovers the specific original limit:

$$L = \frac{1+1/2}{1-1/2} = 3. \quad (7)$$

□